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Bridging critical institutionalism and fragmented authoritarianism in China: An analysis of centralized water policies and their local implementation in semi-arid irrigation districts

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Abstract

This paper attempts to bridge critical institutionalism and fragmented authoritarianism in China. Moving beyond the predictive functionalist perspective and the overemphasis on state primacy, the paper focuses on how top-down water policies are exercised by local agencies, collective communities, and individuals at the county level and below. Through a case of irrigation management in the upper reaches of the Yellow River, this paper posits that institutional bricolage and fragmentation are prevalent despite the imposition of the authoritarian state. Moreover, it shows that local water policy implementation is reshaped by various contested discourse, the pursuit of different interests, social relations, and broad transformative forces. This high degree of diversity not only manifests an increasing institutional flexibility, as water policies are operated outside China's hierarchical bureaucratic system, but also is embedded in an era of socio-economic transformation where the processes of industrialization, urbanization, and neo-liberalization have changed the dynamics of rural governance.

Keywords: China, critical institutionalism, fragmented authoritarianism, institutional bricolage, water governance.

1. Introduction

In their article highlighting the scholarly trend of studying institutional bricolage and, more generally, critical institutionalism, Cleaver and de Koning wrote "There is a need to place local institutional arrangements within the wider frames of governance (the political ecology and political economy) which shape the possibilities for resource allocation, adaptation and negotiated solutions," (Cleaver & de Koning 2015, p. 2). Their argument echoes the increasing attention paid by common-pool resource (CPR) scholars to the complexity and uncertainty nested in particular social-ecological environments (Clement 2010; Hall *et al.* 2014). The impacts of contextually sensitive features (i.e. social relations, politics of policies, community heterogeneities, power dynamics, and historical trajectories) on the performance of CPR governance have yet to be fully examined in diverse empirical settings, which therefore raises questions about whether the focus of CPR theory on efficiency and functionality is an oversimplification of complex local conditions (Saunders 2014).

For this line of inquiry, the broad governance context of China, which offers a number of unique socio-political and economic characteristics, has been insufficiently studied in the existing CPR literature. Traditionally, the Chinese political system is characterized by a number of authoritarian weaknesses, such as overcentralization

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of decisionmaking, overreliance on coercive regulations, the prevalence of personal power, and the absence of the rule of law (Nathan 2003; Beeson 2010; Yan 2011). However, it would be overly simplistic to categorize China as an exemplar of authoritarianism and neglect the complex, multivariable, and multi-level institutional and social interplay nested in an increasingly diversified context (Lo 2015). As the pluralized dimensions of the Chinese society unfold, the notion of "fragmented authoritarianism" first proposed by Lieberthal and Oksenberg (1988) has increasingly been used to analyze mixed and malleable policy outcomes shaped within an authoritarian state (Mertha 2009; Nickum 2010; Moore 2014a,b). In this sense, very few scholars have bridged fragmented authoritarianism with critical institutionalism in their studies on institutional dynamics of local CPR governance. Cleaver and de Koning's (2015) question of how institutions are animated by people in particular spaces has yet to be examined in the fragmented authoritarian political regime of China.

Using water governance as an example, this paper stands at the intersection where China's top-down political inputs interact with indigenous social-ecological and economic conditions to scrutinize the process of CPR governance. This paper investigates how state-centric control over water is diversely exercised in rural areas where economic pursuits and social-historical relations at the lowest operational level inflect power relations. This approach does not evaluate the robustness of Ostrom's (1990)design principles with selected cases (Cox *et al.* 2010); instead, it probes the diverse local practices and governance outcomes by moving beyond the predictive functionalist perspective. The operational roles of collective communes and individuals who exercise top-down policies with some degree of "localism" or "self-interest" are effectively included in the analytical scope of critical institutionalism and fragmented authoritarianism. Our objective is to conduct a local-level analysis within the broad political economy and explain a miscellaneous set of local water governance phenomena that emerges in the Chinese context.

In the following sections, we first review the relevant literature on critical institutionalism and fragmented authoritarianism. Next, we introduce a case study of water governance in semi-arid Northwest China. We then elaborate the diverse causes and processes of water policy implementation by probing into local irrigation practices that are entangled in the interplay between central policies, regional physical and social conditions, rural institutions, and politics at the village level. We conclude with a discussion on the implications of the micro-examination of critical institutionalism and fragmented authoritarianism in the Chinese context.

2. Literature review

2.1. Critical institutionalism

Critical institutionalism is as much a critique of as a complement to mainstream institutionalism (Cleaver & de Koning 2015; Ingram *et al.* 2015). Mainstream institutional thinking, signified by the seminal and influential work of Ostrom (1990, 2009, 2010), holds a predictive perspective regarding institutional dynamics and collective action (Johnson 2004). Its underlying presumption is that favorable outcomes (i.e. sustainable CPR use) can be achieved through amendable institutions, which may not only share commonalities in nature but also may be crafted, shaped, and replicated in desirable directions and different scenarios (Hall *et al.* 2014). Mainstream institutionalism has been the subject of a prominent line of inquiry for understanding and crafting long enduring arrangements for governing CPR in both academic and practitioner communities. Critical institutional thinking, although still nascent, often questions the rational choice assumption of the predictive perspective and queries the "fuzziness," complexity, and uncertainty of diverse empirical settings where the explanatory power of mainstream institutionalism seems to be limited (Cleaver & de Koning 2015).

One should note that critical institutionalism does not fully reject mainstream institutional thinking. In fact, their views may often overlap. For instance, both critical institutionalism and mainstream institutional thinking agree with the importance of institutions. Mainstream institutional scholars are also antagonistic to the idea of panacea and pay close attention to the context, scale, and interdependencies when applying Ostrom's design principles (Ostrom *et al.* 2007; Baggio *et al.* 2016; Schlager 2016). Some empirical cases that are inconsistent with the predictions of the design principles have been identified and explained (Barnett *et al.* 2016). Yet the salient distinction is that critical institutionalism is built upon an alternative assumption about the nature of human behavior. It emphasizes that the social, historical, and anthropological dimensions of human interactions are as important as the monitoring, incentives and sanctions that regulate how people individually or collectively use

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the commons. From a critical institutionalism perspective, the practical approaches that people adopt to manage CPR are not crafted or imported but are constantly modified, invented, and restructured, based on whatever resources are available, to adapt to changing challenges and new configurations of rules, as well as transforming norms and socio-economic relationships that are occurring in a particular governance regime. Moreover, these changing institutions could become functional and enduring as they draw legitimacy from accepted knowledge, culture, and established mechanisms of authority. Cleaver (2002, 2012) and de Koning (2014) summarize this dynamic process of ad hoc evolution and legitimization as "institutional bricolage."

The focus on "bricolage" substantially expands the scope of analysis on the commons. Critical institutional scholars are concerned about the multi-faceted and multi-scalar complexity of social interactions embedded in everyday life: micro-politics, historical formation, and biophysical geographies; therefore, they are skeptical about whether institutions can be mechanically crafted to be efficient and whether institutional interventions, such as participation, empowerment, and social learning can be effectively translated into workable arrangements under complex empirical conditions. In this sense, critical institutionalism embraces a larger pool of literature than mainstream institutionalism. The focus of critical institutionalism regarding the transformation of institutions, the operationalization of institutions by human beings, and the relationship between institutional form and outcomes requires a comprehensive examination of institutional dynamics and their compatibility with local conditions (Leong & Lejano 2016).

The impacts of broader governance on local institutions are one of the key themes of critical institutionalism (Cleaver 2012; Cleaver & de Koning 2015). This line of inquiry could draw insights from political economy and political ecology analysis, which focuses on unequal and dynamic power relations as well as their roles in shaping institutional arrangements for CPR governance at the local level (Blaikie 2006; Andersson & Ostrom 2008; Jones 2015). As a fragmented authoritarian regime under rapid socio-economic transformation, China offers a valuable case for assessing the process of "institutional bricolage." With very little research placing critical institutionalism in the broader political context of China, this paper aims at engaging in a discussion with the CPR literature through a Chinese case of local irrigation governance arrangements, which are developed in the dynamic mixture of externally imposed institutional crafting, as well as socially and historically embedded interactions.

2.2. Fragmented authoritarianism

As a widely used and durable analytical framework for understanding Chinese politics in the open and reform era, fragmented authoritarianism captures the key features of the evolution of the Chinese political process, during which the state's dominance has been gradually eroded by the pluralization of civil society and the development of the market, while the state's power remains strong enough to maintain its political and socio-economic agenda against demands from non-state actors (Mertha 2009). In the field of environmental governance, two main streams of scholarship have been developed under the framework of fragmented authoritarianism. One stream focuses on policymaking processes during which previously marginalized actors, such as non-governmental organizations (NGOs), the media, and individuals, seek to, or even manage to, insert their concerns into centralized policies (Mertha 2009; Grabosky 2013; van Rooij *et al.* 2016). The other stream focuses on policy implementation processes where problems of opportunistic behavior, poor coordination, and weak enforcement capacity in local states are prevalent (Feng *et al.* 2006; Marks 2010; Ran 2013). In this sense, "fragmentation" could be a trend toward pluralization – the incremental inclusion of diverse non-state actors in the policymaking process, in addition to a closed circle of party and state officials, seems to be a positive response to diverse demands from society (Lema & Ruby 2007; Zhang *et al.* 2013a; Shen 2014). Additionally, "fragmentation" could be formed by cross-level and cross-sectoral bargaining and variant operational contexts in which the diverse political goals of the vertical strands of functional agencies (*tiao*) and the horizontal threads of territorially based administrative units (*kuai*) are mixed and reshaped, leading to weak enforcement and even the failure of centralized policies at the subnational level (Marks 2010; Kostka 2016).

After a careful review of the current literature on the entry of "new actors" in policymaking processes and the prevalence of "command-without-control" in policy implementation, a salient gap has emerged. This gap points to an inadequate examination of the roles of local practitioners, which mainly include county governments and lower level agencies, as well as rural communities and peasants, who exercise many of the top-down policies. In

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other words, it is problematic to study China's cross-scale political interplay within only the bureaucratic system or simply from a mechanical perspective. Given the growing complexity of the social reality, one should recognize that a full circle policy process involves continuous interactions among different groups of actors whose divergent interests, strategies, and reasoning are ineluctably intertwined from policy formulation to policy operation. As Rawski notes, "official policy is only part of the reform story (Rawski 1995, p. 1169)." It is no longer news that local states do not passively accept everything externally imposed on them, and they do not fully control how policy outcomes are translated in empirical settings (Olivier de Sardan 1988). In this sense, local states are not "a cog in the machine" that only enforce state-centric decisions at the end of the policy process; rather, they could be a starting point at which new rules might be developed in specific local contexts and diversely operated by non-state actors (i.e. peasants) in their daily lives.

2.3. Bridging critical institutionalism and fragmented authoritarianism

Although derived from different fields of study, critical institutionalism and fragmented authoritarianism converge at dynamic local spaces where complex, uncertain, and evolving networks of social relations, micro-politics, and economic transformation capture the attention of both lines of scholarship. Through the lens of irrigation management in a Chinese context, the integration of critical institutionalism and fragmented authoritarianism raises three strands of key questions that warrant further examination.

First, can centralized water policies initiated by the state at the national level be effectively enforced at the local level? To what extent do the practices derived from these policies represent the state's aspiration for water governance reform? Is there evidence of "bricolage" that reshapes policy implementation? This strand of questions probes into the interfaces between authoritative power and its exercise, between national discourse and rural values, and between regulation and practice. It is of great importance to examine the translation of power across the hierarchical order and the occurrence of institutional changes that characterize the fragmentation of China's authoritarian political system (Yang 2013; Lo 2015; van Rooij *et al.* 2016. Likewise, critical institutional scholars also often focus on these interfaces where bricolage emerges to navigate discrepant logic, knowledge, values, and pursuits of interests (Cleaver & de Koning 2015; Funder & Marani 2015; Marin & Bjorklund 2015).

Second, to what extent do imported institutions, which represent examples of "best practices," fit with local conditions? How can we understand the emergence of locally driven rules-in-use through a process of institutional bricolage? Against the external imposition of institutional crafting, can innovations arise from adapting particular contexts and improvising on existing practices? For institutional scholars, the concept of institutional fit is intuitively appealing, as it associates crafted institutions with various dimensions of the social-ecological system with which they interact (Young 2002; Folke *et al.* 2007; Epstein *et al.* 2015). However, institutional fit is also a relatively vague concept that leaves unanswered questions in regard to the defining features of fit and the typology of institutional arrangements for facilitating fit across scales and levels (Cox 2008; Vatn & Vedeld 2012). The complexity and uncertainty embedded in different social-ecological systems make it difficult to apply "best practices" in a consistent way. In this sense, the growing discussion about adaption, resilience, social learning, and vulnerability suggests an alternative perspective of "fit" (Pahl-Wostl *et al.* 2007; Carpenter *et al.* 2009; Young 2010; Folke *et al.* 2010, 2016). This alternative perspective essentially acknowledges the generality of "misfit" and calls for a realistic approach rather than an idealistic attempt to import "best practices" for governing CPRs (Keskitalo *et al.* 2016). The alternative view is not only consistent with emphasis on the sensitivity and legitimacy mirrored in institutional bricolage but also provokes discussion of the literature on fragmented authoritarianism in China, as it concerns the robustness of an authoritarian state and the sources of its authoritarian resilience under both internal and external challenges (Perry 2007; Cai 2008; Schubert 2008; Yan 2011). The frameworks of responsive authoritarianism (Grabosky 2013; Chen *et al.* 2016) and consultative authoritarianism (He & Thogersen 2010; Teets 2013) have been developed to analyze policy innovations, institutional reforms, and polycentricism in China, which are presumed to be excluded from authoritarian regimes. The dynamics of fragmented authoritarianism, along with the emergence of local institutions during the process of bricolage, jointly offer a context where new organizational forms of institutions can be studied.

Finally, what are the impacts of socio-economic and demographic changes on local institutional arrangements? How are these changes associated with the rapidly evolving Chinese political economy? Unlike previous

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focus on the translation of power and the emergence of local institutions, the last strand of questions places extra attention on societal changes in rural communities during China's rapid economic development. This is a direct response to Cleaver and de Koning's (2015) argument of the role of a broader political economy in shaping institutions. In particular, the massive scale and breakneck speed of economic development, which is led by the authoritarian state and facilitated by a pluralizing private sector and civil society, have substantially altered the socio-economic structure of Chinese rural communities. Rural–urban migration, urban land expansion, diversification of agriculture, and improvement of productivity have characterized a dynamic environment of institutional changes. In contrast to a relatively static environment in Southeast Asia and Latin America, where many previous studies of irrigation management have been conducted, the transforming Chinese political economy contains a unique "recipe" for vigorous bricolage and fragmented authoritarianism that have yet to be fully investigated by institutional scholars.

3. Case and method

Water control has been instrumental in pushing forward the state's political, economic, and social agendas in China. From the Maoist era, when agricultural hydraulic constructions were associated with the establishment of rural collectives and the mobilization of rural labor (Perkins 1969; Li & Yang 2005, to the open and reform era, when the Chinese state carried out a number of large-scale water projects (i.e. Three Gorges Dam and the South-to-North Water Diversion Project) to maintain rapid economic development (Webber *et al.* 2017), the dispersed practices of water governance in China have been inescapably political both in and beyond the state (Agnew 2011; Rogers *et al.* 2016). Water governance comprises interactions not only within the bureaucratic system at various levels but also among different sectors of society, where enterprises, activists, NGOs, the media, and individuals are more or less involved in the process. These complexities of China's water governance system make it a suitable case for a study of how institutional bricolage and fragmented authoritarianism have evolved in an increasingly diversified context.

The Ningxia Hui Autonomous Region (Ningxia) is selected as a case study to facilitate a detailed elaboration of the dynamics of bricolage and fragmentation nested in China's water governance system. Because Ningxia is located in a semi-arid region, it contains a broadening gap between water supply and demand, which vividly illustrates China's nationwide water scarcity problem. With the Yellow River flowing across its territory, Ningxia also encompasses a sophisticated irrigation system with a long agricultural history and a number of innovative water institutions. The variations in irrigation and agricultural activities could offer a valuable opportunity and rich material to examine how water governance is exercised at the local level.

Most of our fieldwork was carried out in two county-level regions, namely, Qingtongxia (QTX) and Hongsibu (HSB). We conducted fieldwork in eight villages located within QTX and HSB to facilitate a detailed examination of the local operation of water governance (Fig. 1). Although these case study sites are all located within the Wuzhong municipality, they demonstrate distinctive features of irrigation. QTX is close to water sources, is located in a flat flood plain, and adopts a traditional irrigation system in which surface water is distributed by gravity. However, the gravity irrigation system cannot reach HSB, which is located approximately 50 km away from the mainstream of the Yellow River and 100 m above the surface water level of the nearest main irrigation channel. A series of pumping stations and channels have been built since the 1990s as part of a poverty alleviation project to artificially lift water to HSB, one of China's most impoverished areas. These divided irrigation methods are an adaptation to different geophysical conditions and have been formed by heterogeneous social and agricultural contexts.

Data are mainly drawn from semi-structured interviews and group discussions that were conducted in four field trips to Ningxia from September 2014 to January 2016. Each field trip lasted approximately two weeks. During these field trips, we first approached the Water Resources Department of the Ningxia provincial government, where group discussions were held with seven government officials with intimate knowledge of Ningxia's water governance issues. Various topics ranging from water consumption control to institutional innovation (e.g. water rights transfer) were discussed. At our request, we were then introduced to the Water Bureaus of the QTX and HSB governments, where we conducted individual interviews with six officials directly involved with the implementation of top-down water policies at the county level and below. With their help, we interviewed

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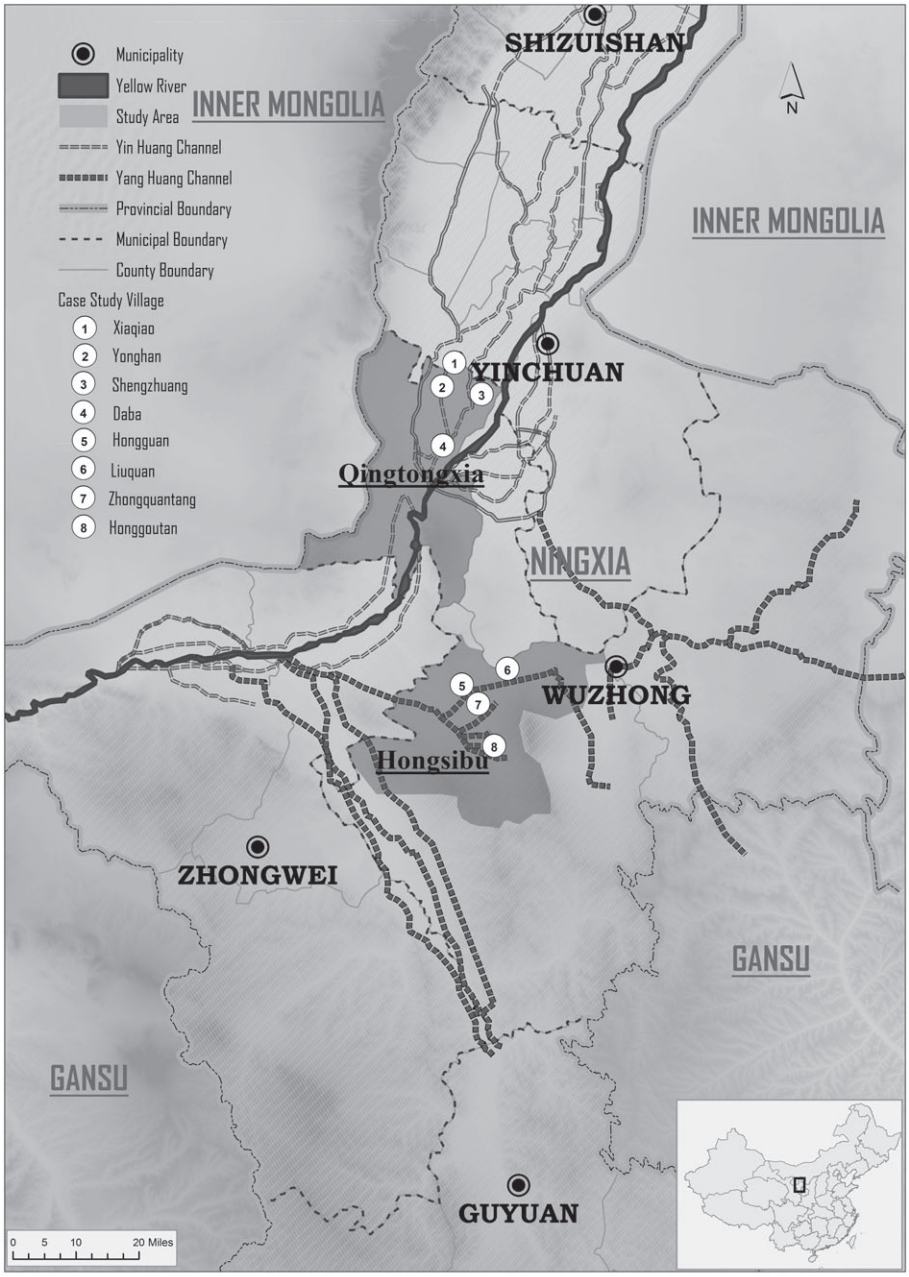

Figure 1 The case study area.

each and every chairman of the local Water User Associations (WUAs) in the eight case study villages, which were selected to cover different biophysical conditions in both gravity and lift irrigation districts. In addition, we hosted group discussions in each village with representatives of the village committee, who are core members of the WUAs. The topics of the interviews and group discussions at the village level included basic demographic and socio-economic characteristics of the village, land use and agricultural activities, irrigation arrangements, water conflict and resolution mechanisms, external policies and their local enforcement, village history, social relations, and so forth. We also collected information regarding these topics through two random household visits in each village, as well as through interviews with three scholars from the Ningxia Academy of Agricultural Science who had previous fieldwork experience in these villages. The results corroborated what we learnt from the interviews and group discussions. We did not conduct a large-scale household survey, mainly because of the high degree of homogeneity among different households in terms of their agricultural and socio-economic

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characteristics within each village. In addition, this qualitative study does not aim to establish causal relations between variables based on statistical models, but to explore complexity and diversity in institutions. Therefore, it does not need a representative sample of respondents, but those who are qualified to present variety in experience, beliefs, and opinions about irrigation institutions and rural governance. We believe our research methods have served this purpose. Secondary data, such as official documents, project reports, and academic publications, were also used as supplementary sources to assist the analysis.

4. Analysis

Previous research on China's hydro-politics and policy implementation has primarily centered on central-local relations within the formal political system, such as compliance with mandates, cadre management and assessment, and corruption (O'Brien & Li 1999; Cai 2004; Hillman 2010; Smith 2010; Zhou 2010; Ahlers & Schubert 2015). Alternatively, we focus on the prosaic dimensions of the policy implementation process and the roles of rural communities that eventually implement such policies and incorporate them into their everyday lives, despite the fact they are not formal actors. With the objective of examining the process of "bricolage" and "fragmentation" through the lens of irrigation management, we reveal the dynamics of institutional changes from three aspects, namely, the exercise of the state's power, the import of public participatory organizations and the emergence of local institutions, and the impacts of socio-economic changes. We approach this task by describing the irrigation management practices and local socio-economic activities in two groups of villages located in the gravity irrigation districts of QTX and the lifting irrigation districts of HSB.

4.1. The exercise of power

Water governance in China demonstrates prominent features of command-and-control, state-centric planning, and technocracy (Wang *et al.* 2017). This governance system is based upon the 1982 Constitution and the 2002 Water Law. The former established the public ownership of all natural resources, including water (Article 9), and the latter established an overall framework for both river basin agencies and territorially-based administrative agencies to oversee various water governance issues, including water planning (Article 17), water allocation (Article 44), gross volume control, and ration management (Article 47). Under this overarching governance framework, the Chinese state seems to be the sole dominant actor in the arena of water governance. We have identified three salient approaches regarding the state's control of water in our case study area.

First, the Yellow River water allocation plan, which is imposed by the state on lower-level jurisdictions (including Ningxia) through top-down administrative orders, divides limited water resources among different jurisdictions, sectors, or industries (State Council 1987; Shen & Speed 2009). In principle, each water user must acquire a water extraction permit and is not allowed to exceed their water-withdraw allotment. Second, the state's control of water consumption is further enhanced by its monopoly of the water supply. In the irrigation districts of QTX and HSB, the peasants, as water users, must purchase water from the suppliers who primarily include channel management agencies and pumping stations. Both water suppliers are public institutions affiliated with the Water Resources Department of the provincial government. Their operation is therefore subject to scrutiny by their superordinate, whose primary responsibilities include keeping regional water consumption within the allocated quotas. Last but not least, the promotion of new technologies is another main policy instrument for regulating water consumption. In addition to the top-down imposed regulations, the state has provided funding support for a water-saving infrastructure for peasants, with the objective of facilitating local adaptation to stringent water conditions.

However, it is overly simplistic to conclude that these state-centric arrangements can effectively prevent the overexploitation of water resources. Authoritarian power seems to be dwindling in the process of policy implementation at the micro level.

Particularly because a top priority in cadre assessments is to maintain social stability (Lee & Zhang 2013), the local government definitely does not want to see collective rural resistance caused by cutting off the water supply to peasants. An official from the Provincial Water Resources Department stated:

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> We have talked about demand management for quite a long time, but in practice, our main tasks remain to ensure water supply. Neither the government nor the peasants can afford the losses from water shortages.[2]

In fact, none of the eight villages we visited had experienced water suspensions resulting from constraints on the water allotment; on the contrary, when shortage problems occur during busy times of the year or dry years, ad hoc arrangements for provisional water supply, such as temporary water diversion and the allocation of contingent allotments, are usually made to ensure agricultural production and food security. As a consequence, the "red line" of water consumption seems crossable when it mingles with other governance issues. This seemingly improvised result, from the bricolage perspective, revealed a hidden logic of institutions; namely, water consumption control is yet to be fully accepted in practice as it is contested by dominating discourses of social stability and livelihood improvement.

Second, the water suppliers, as public institutions, were reluctant to reduce water consumption because they receive significant income from selling water to the peasants. The staff salaries of channel management agencies were not fully supported by budgetary appropriations until 2013, so the collected water fees had served as extra-budgetary income to support the agencies' operation.[3] Under this arrangement, corruption and opportunistic behavior occurred among the water suppliers. A few peasants from HSB revealed:

> Last year, our villagers had a fight with the people from the pumping station. Our village is located at the end of the channel, so it is difficult for us to get enough water. But we found out that the pumping station leased out a piece of land nearby to an outside investor and secretly sold water to the investor. They made individual profits from public services, and it cost us badly.[4]

In this sense, the state's control of the water supply is not necessarily effective in curtailing water consumption, although the water suppliers are supposed to fulfill their public duties by enforcing governmental regulations.

Finally, the promotion of the water-saving infrastructure has also encountered some difficulties because of the lack of social support. Some peasants are skeptical about the effectiveness of drip irrigation, as it contradicts the conventional wisdom of "the more water there is, the better crop yield." One of the local peasants expressed his concern:

> I know this is technology, but there must be some impact[s] if less water flows into my farmland. Would the government compensate my losses if I could not produce as much as I did before?[5]

Other peasants have concerns regarding the continuity of financial support. The current policy for supporting the water-saving infrastructure is a temporary approach. There is no clear timeline in regard to how long the support will last. The peasants are reluctant, as they do not wish to bear the costs if fiscal transfer from the central government stops.

This analysis shows that seemingly strong water governance institutions are reshaped by contested policy discourses, opportunism, and the lack of social understanding of new technologies, demonstrating that there is evidence of fragmentation and bricolage during the process of power translation when the authoritarian state's water control policies are diversely exercised in rural communities.

4.2. The import of Water User Associations (WUAs) and the emergence of local institutions

In addition to the domestic water institutions imposed by the authoritarian state, externally imported institutions have been crafted in our case study area. WUAs, which are considered an effective public participatory approach for maintaining the sustainable use of water resources (Wang *et al.* 2007; Zhang *et al.* 2013b; Hu *et al.* 2014; Huang 2014), have been widely established since the early 2000s in conjunction with World Bank projects at the requirement of the Ningxia government (Wang *et al.* 2010). Each of our case study villages has a WUA, which is mainly responsible for managing various local water affairs, such as the collection of water fees, the organization of irrigation activities, and the mobilization of collective action for maintaining irrigation facilities. However, the efficacy of the WUAs is unexceptional.

In practice, we found a substantial organizational and functional overlap between the WUAs and the village committees; the latter is the main village organization for attending to public affairs in rural China. Thus, the

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establishment of WUAs does not change the underlying structure of village governance. The reason is that water issues are not always about water per se; irrigation management often involves the coordination of collective action, the reconciliation of tensions, and the balancing of interests among villagers. This situation means that water issues cannot be isolated from other collective matters, which often fall outside the jurisdiction of the WUAs. According to a WUA chairman:

> A villager might refuse to pay water fees because he felt that he was unfairly treated last year when the village appropriated a piece of his land for road construction. He demands more compensation; otherwise he [will] not pay the water fees.[6]

This type of resistant action is not a rejection of the concept of "paying water fees as a duty" but a strategy to pursue his belief of fairness in a rural community where various collective affairs are interconnected and thus more complicated than water per se.

Furthermore, the operation of the WUAs is exceptionally challenging in a closely connected Chinese rural society where "everyone knows everyone." The enforcement of rules and sanctions is challenged by blurry boundaries between "right" and "wrong." For instance, it would seem callous, and practically unfeasible, to punish those who violate rules, such as those who pay their water fees late, those who commit petty water thefts, or those who are "free-riders." Sanctioning is even more impracticable when the majority of the violators did not intentionally break the rules. Their violations or failures might be attributed to other socio-economic reasons, such as large medical expenditures, delayed salary payments, or the loss of the ability to work as a result of injury or sickness. Under these conditions, strict punishments (i.e. water supply cutoff) would have caused greater economic losses to already vulnerable households. In practice, bending the rules to accommodate unique situations in rural communities is not considered "corruption;" rather, both the WUAs and water suppliers have to adjust, reshape, and even improvise the rules so that they are more compatible with the multifarious local social conditions. In other words, many functions of WUAs have not been legitimized in the rural Chinese context. The sanctioning rules, as externally imported institutions, are contested and even rejected by a number of endogenous social norms, which accommodate the sense of reciprocity and kindness rather than rigid institutional enforcement at the community level.

Although the impacts of WUAs seem limited, some locally initiated institutional innovations have emerged in our case study area. Water rights reform is a salient example. China has a complex water rights system as a result of a mix of ownership (e.g. collective-owned and state-owned), a mismatch between water suppliers (those who withdraw from water bodies), and water appropriators (those who consume water resources), as well as instable water availability in terms of both quantity and quality. In general, state ownership of water resources is exercised by the State Council and delegated to different subnational governmental agencies through water allocation plans. The right to withdraw water is given to a variety of water suppliers (i.e. pumping stations) who obtain permits from water agencies.[7] The right to utilize water is enjoyed by a large number of stakeholders, such as irrigation districts, communes, peasants, urban residents, and manufacturing industries who purchase water from water suppliers.

In this complex system, local attempts to initiate new irrigation rules often involve innovative interpretations or applications of unclearly defined water rights. For instance, Yanchi County, another county located in Wuzhong municipality, redistributes all of its water allotment all the way down to the household level. The household irrigation allotments are delineated according to the area of farmland and farming pattern, which essentially combines the water use right with the land use right. Each household's allotment is documented by a water ticket system issued by the local water management agency. The peasants have to present a water ticket each time they need water, but the ticket cannot be traded. This is an approach that reallocates public water rights outside of the administrative matrix, which demonstrates private characteristics but has yet to be clarified by the legal framework. In HSB, an online platform through which water use allotments can be traded was established within the county. This platform is only accessible to a small number of main water users, but it is a more audacious attempt at water reallocation. Unlike the water ticket system, which only gives rural households "the right to use" the water, the platform allows the "right to benefit" from the water, which represents a higher level of private right and thus requires more legal and political scrutiny in the socialistic regime.

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Although there is a limit to the scope and efficacy of these new policies, they represent a local solution to local problems. These policies were neither imposed by the state nor imported from "best practices" but emerged from empirical local needs. This phenomenon demonstrates strong Chinese characteristics of policy innovation, which draws upon experiences of problem solving at the local level (Heilmann 2008). The "grey area," where unclarified property rights and unorthodox practices are allowed, offers an opportunity for different interpretations of institutions, and thus may contribute to the fragmentation of policy outcomes in local practices. A local official advised:

> Our allotment is not enough, but we need to develop. We are a poor county in a developing country. People want to live a better life by farming more and making more money. My superiors, on the other hand, need to enforce the gross volume control target. Therefore, we must figure something out to accomplish both. We use the water tickets to let the peasants know that water is not enough, but we also try to ask for more water use quotas for more development.[8]

Similarly, a local official from the agricultural department shared his opinion as to why institutions cannot be mechanically implemented or imported in rural areas:

> The peasants are very simple. Their concerns are mostly about money. However, the peasants are also complicated. There is a complex social network that cannot be easily changed or understood by outsiders.[9]

4.3. The impacts of socio-economic changes

Although Chinese rural society maintains a strong sense of community and social cohesion as well as unique local characteristics, it has been rapidly transforming since the 1980s (Guo 2001). The implementation of the household responsibility system has substantially increased agricultural productivity and freed hundreds of millions of rural laborers (Krusekopf 2002). China's continuous economic reforms have proceeded at a rapid pace toward urbanization and modernization, which have contributed greatly to rural–urban migration and structural changes in rural governance (Chan & Zhang 1999; Liu & Murphy 2006).

The socio-economic changes associated with China's rapid development have led to significant impacts on irrigation institutions. For instance, we witnessed a trend toward the marginalization of irrigation in rural affairs. A WUA chairman advised that it was very difficult to find villagers willing to work in the WUA:

> It is simple math. A young man could make at least 100 Yuan a day working construction or manufacturing jobs in the city nearby, but working as a WUA staff member pays only slightly more than 1,000 Yuan a month. Sometimes when we cannot collect water fees from the villagers, we have to use our own money to fill in the hole.[10]

In this sense, the lack of financial incentive also explains the functional overlap between the WUAs and village committees. With young and capable labor pursuing a better life in the cities, village cadres have to remain as chairmen and members of the WUAs who attend to irrigation and other collective affairs together.

In addition to the marginalization of irrigation, changes to land policy, the agricultural paradigm, and their induced social-political problems have also influenced the irrigation method (i.e. applying a water-saving infrastructure) and the change in relevant water institutions. A salient example is the promotion of land transfer and modern agriculture. At the national level, this is a policy response to increasing rural–urban migration and relatively low agricultural efficiency caused by farmland fragmentation.[11] The policy allows peasants to transfer their land-use rights (gained from the collective) to other farmers or economic entities (Ye 2015). In other words, the migrants, who mostly work in the cities instead of on farmland, could lease out their contracted land to large land holders, such as family farms, peasant cooperatives, and even investors. As a result, fragmented farmland plots can be integrated, which promotes agricultural commoditization, specialization, and modernization, including the establishment of a modern water-saving infrastructure. However, scaling up farmland is not possible unless a reasonably large number of peasants with adjacent farmland all agree to the transfer. Many social conflicts arise during the bargaining and operation of land transfers (Gong & Zhang 2017), which then impede the application of water-saving infrastructure that requires relatively large-scale agricultural production. In this sense, the barriers

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to effective irrigation management are not directly water-related; they could instead be a result of socio-economic changes.

Socio-economic development in China also brings about changes to peasants' attitudes about agricultural activities and irrigation. With fading differences between urban and rural lifestyles and increasing living standards for peasants, the extra work associated with the maintenance of new irrigation infrastructure now seems burdensome. A local peasant told us:

> Drip irrigation is a good thing. I could control the amount of water. It is good for my grape farm and saves water. A few years ago, the government gave us money to build a small water tank for drip irrigation. Many of us tried but eventually gave up. The Yellow River water contains too much silt, it clogs the dripping pipes, and we must clear the sediment in the water tank regularly. This is too much trouble. Additionally, we do not want to be tied here to the farmland. Now the water tank is abandoned.[12]

We cannot assert that the failure of the drip irrigation case is typical. However, we argue that new collective action and social problems related to irrigation institutions stem from socio-economic changes in the broader Chinese economy. The outcome of water governance in a particular region could be influenced by the changes in agricultural patterns and population, the expansion of farmland area, the transfer of farmland and water rights, and the price fluctuation of agricultural products, which warrant further examination in future studies.

5. Discussion and conclusions

Water governance in China has rapidly and profoundly changed over the past decade. This process has been guided by the state's aspiration for maintaining sustainable water use, improving water use efficiency and optimizing water distribution among regions, sectors, and individual users. Meanwhile, we have witnessed a variety of local actions resulting in fragmentation and bricolage in this seemingly authoritarian reform. The combination of unification and decentralization, state-centrism and localism, and top-down and bottom-up jointly makes water governance an interesting case for the study of critical intuitionalism and Chinese politics.

Using water governance in the Ningxia irrigation districts as a case study, this paper engages in a discussion of critical institutionalism and fragmented authoritarianism with empirical evidence encapsulated in three dimensions, namely, the exercise of power in controlling water, the import of the WUAs, and the emergence of local water rights institutions, as well as the impacts of socio-economic changes on collective action. First, we have shown that the power of the state in controlling the water supply has been reshaped during its translation by contested policy discourses, the pursuit of different interests, varying social norms, and diverse relations in rural context. Rather than focusing on the principal agent problem between central and local states, we have demonstrated that the exercise of authoritarian power is fragmented outside the bureaucratic system where diverse social-political and historical elements play a key role in influencing policy outcomes. Second, the insights generated from the examination of "fragmentation" in an authoritarian context demonstrate that the process of "institutional bricolage" is prevalent in irrigation practices. Not only do local actors adapt to the state's control of total water consumption, their actions against externally imposed policies have also been legitimized by established social norms. The persistent function of village committees indicates that local irrigation practices are still dominated by social-historical relations rather than imported institutions such as WUAs. Moreover, problem-solving-oriented local practices even stimulated diverse institutional innovations (i.e. water rights reform), which simultaneously respond to the state's objective of water consumption control and the farmers' desire for benefit maximization. Last but not least, the socio-economic changes associated with the macro-level political economy of China have substantially impacted local irrigation practices and institutions. The rapid urbanization and modernization processes have led to changing demographic and social conditions in rural communities. These changes, in turn, raised new collective action and social problems that enhance fragmentation and bricolage.

Our paper examines the implementation of water policies in a rural context, where a variety of institutional and socio-economic conditions are manifested in local irrigation practices. In contrast to investigating how local states' actions are controlled by the central government, our approach focuses on the extensive variations in the seemingly ad hoc ways through which water resources are governed at the grassroots level. This is an attempt to analyze diverse policy outcomes by incorporating the multifarious aspects of rural society into an authoritarian

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water governance system. Although the impacts of the varied degrees of coordination, compartmentalization, or segmentation throughout the bureaucratic system have been documented (Mertha 2009; Moore 2014a; Nickum & Lee 2006; van Rooij *et al.* 2016), our findings suggest that policy outcomes are not solely tailored by formal institutions. During implementation, some policies are accepted, some are reshaped, and some are rejected. It is the extent of compatibility, synergies, and tradeoffs between external and indigenous forces and between formal and informal arrangements that have jointly shaped the intricate patterns of local arrangements and practices. In this sense, water governance in a rural context is not only framed by authorities, institutions, and their internal interactions but also embedded in local social and cultural conditions (Cleaver 2012; Cleaver & de Koning 2015).

Our paper intends to avoid overemphasis on the outsized influence of the Chinese state (Gilley 2012) or crafted institutions and moves beyond the dichotomous division between democracy and authoritarianism, between effectiveness and ineffectiveness, and between "command-and-control" and "command-without-control." Using the case of irrigation management, we probe the malleable water policy outcomes of an authoritarian system by focusing on local exercises at the county level and below. The focus on bricolage and fragmentation is not a rejection of mainstream institutionalism or the overarching authoritarian context but an exploration of a myriad of improvised rules, local pursuits of interests, competition, tradeoffs, and adjustments, which constantly reshape the outcomes of policies during implementation. In light of the prevalence of bricolage and fragmentation, we argue that a more nuanced understanding can be generated from the examination of flexible local practices than from a simple evaluation of regulation compliance or the test of imported "best practices." Diverse policies and outcomes are produced by a combination of institutions, socially embedded relations and broader political-economic conditions, which are manifestations of the agenda of the state, the power structure, local heterogeneities, and the transforming economic context. The ways and the extent to which these diverse outcomes are shaped warrant more scholarly attention. For instance, one would be intrigued to investigate the relationship between authoritarianism and bricolage and to understand why some of these institutions are legitimized and even promoted by the authority while others are rejected. In this sense, more contextually sensitive research on critical institutionalism and fragmented authoritarianism could shed light on some previously disregarded details that might account for these diverse policy processes and malleable policy outcomes in the authoritarian Chinese state.

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Notes

1 Interview with a local-level official, HSB, November 2015.
2 Group discussion at the Provincial Water Resources Department, Yinchuan, October 2015.
3 Group discussion at the Provincial Water Resources Department, Yinchuan, October 2015.
4 Interview with local peasants, HSB, January 2016.
5 Interview with local peasants, QTX, January 2016.
6 Interview with local peasants, QTX, January 2016.
7 Industrial and agricultural permit-holders are also allowed to directly withdraw surface or ground water.
8 Interview with a local-level official, Yanchi, October 2015.
9 Interview with a local-level official, HSB, January 2016.
10 Interview with WUAs, HSB, January 2016.
11 The quality of farmland varies in most Chinese villages. Some of the lands are more fertile and closer to water sources, while others are less conducive to agriculture. In fairness to all, each household is given several plots of farmland with

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different qualities, located at different places in the village. Therefore, farmland plots in most Chinese villages are scattered across the village for fairness reasons.

12 Interview with local peasant, QTX, 20 January 2016.

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